Supplementary Material for "A Reproducible ROI-Robust Evaluation and Mobile Distillation Framework for Ultrasound Lesion Classification"

S1. Result Provenance

This supplement retains only tables generated from the final fixed reporting protocol and removes earlier development tables that mixed preliminary data loaders, intermediate exports, and non-final runtime artifacts. The tables below are tied to the strict result folders distributed with the reproducibility package and use the functional model names reported in the main manuscript.

Table 1: Primary result sources used by the current manuscript. Internal run directories are archived in the reproducibility package; the manuscript reports functional model names rather than legacy experiment codes.

Evidence block	Archived source family	Notes
Dataset-specific benchmark	Strict main reruns	TN5000 3 seeds; BUSI/AUL 5 folds
TransXNet-family module sweep	Strict TN5000 module reruns	8 MCA/MUDD/DA combinations
ROI robustness	Strict box-noise and detector-ROI report	GT, noisy, and detector-derived boxes
Mobile export/runtime	Strict TN5000 mobile export and two-device Android reports	Re-exported artifacts
Cross-organ probe	EfficientFormer joint/LODO aggregate report	3 seeds
BUSI duplicate audit	BUSI duplicate audit	Descriptive only

S2. Full Main Benchmark Context

S3. TransXNet-family Design-space Details

S4. ROI Robustness

S5. Android Runtime Details

S6. Multi-organ Joint and LODO Details

S7. BUSI Duplicate Audit

BUSI is retained in its public benchmark form for comparability, but we audited exact and near-duplicate candidates because duplicate or label-

Table 2: Dataset-specific benchmark context under the fixed reporting protocol. Values are mean ± standard deviation over TN5000 seeds or BUSI/AUL training folds evaluated on the fixed held-out test sets.

Dataset	Method	AUC	BalAcc	F1	Acc
TN5000	UL-TransXNet	0.9537 ± 0.0015	0.8865 ± 0.0064	0.8750 ± 0.0059	0.8960 ± 0.0075
TN5000	ConvNeXt-Tiny	0.9508 ± 0.0020	0.8794 ± 0.0023	0.8731 ± 0.0066	0.8957 ± 0.0065
TN5000	Original TransXNet	0.9467 ± 0.0009	0.8727 ± 0.0023	0.8578 ± 0.0120	0.8803 ± 0.0137
BUSI	UL-TransXNet	0.8494 ± 0.0039	0.7702 ± 0.0076	0.7479 ± 0.0283	0.7752 ± 0.0383
BUSI	Swin-T	0.8190 ± 0.0091	0.7137 ± 0.0095	0.6896 ± 0.0300	0.7194 ± 0.0445
BUSI	Original TransXNet	0.8136 ± 0.0239	0.7531 ± 0.0303	0.7441 ± 0.0292	0.7814 ± 0.0304
AUL	UL-TransXNet	0.8146 ± 0.0165	0.7611 ± 0.0283	0.7589 ± 0.0307	0.7827 ± 0.0329
AUL	Swin-T	0.8319 ± 0.0117	0.7783 ± 0.0325	0.7589 ± 0.0243	0.7717 ± 0.0211
AUL	Original TransXNet	0.7899 ± 0.0090	0.7566 ± 0.0456	0.7547 ± 0.0516	0.7795 ± 0.0539

Table 3: TN5000 TransXNet-family ablation with seed variability. Values are mean ± standard deviation over three seeds.

Variant	AUC	BalAcc	F1	Acc
TransXNet-base	0.9456 ± 0.0006	0.8769 ± 0.0014	0.8526 ± 0.0097	0.8733 ± 0.0107
TransXNet-DA	0.9447 ± 0.0024	0.8785 ± 0.0047	0.8732 ± 0.0075	0.8960 ± 0.0088
TransXNet-MUDD	0.9511 ± 0.0018	0.8857 ± 0.0071	0.8785 ± 0.0019	0.9000 ± 0.0022
TransXNet-MUDD+DA	0.9579 ± 0.0002	0.8901 ± 0.0080	0.8836 ± 0.0074	0.9043 ± 0.0063
TransXNet-MCA	0.9465 ± 0.0046	0.8802 ± 0.0068	0.8663 ± 0.0086	0.8880 ± 0.0085
TransXNet-MCA+DA	0.9441 ± 0.0040	0.8747 ± 0.0078	0.8593 ± 0.0094	0.8817 ± 0.0103
TransXNet-MCA+MUDD	0.9555 ± 0.0011	0.8834 ± 0.0013	0.8757 ± 0.0108	0.8973 ± 0.0119
UL-TransXNet	0.9522 ± 0.0009	0.8754 ± 0.0128	0.8652 ± 0.0004	0.8883 ± 0.0033

Table 4: TN5000 AUC robustness context across the four models shown in the main robustness figure under the fixed reporting protocol.

Model	GT	5% noise	10% noise	20% noise	Detector
TransXNet-MUDD+DA	0.9623	0.9594	0.9556	0.9463	0.9497
TransXNet-base	0.9469	0.9445	0.9392	0.9292	0.9388
ConvNeXt-Tiny	0.9534	0.9533	0.9524	0.9483	0.9511
Swin-T	0.9506	0.9509	0.9488	0.9463	0.9481

Table 5: Exact TN5000 ROI robustness metrics for the strongest TransXNet-MUDD+DA teacher.

Input box	AUC	BalAcc	Mean box IoU
GT ROI	0.9623	0.9036	1.000
GT ROI + 5% noise	0.9594	0.8969	0.899
GT ROI + 10% noise	0.9556	0.8842	0.810
GT ROI + 20% noise	0.9463	0.8665	0.662
Detector ROI	0.9497	0.8849	0.789

Table 6: Two-device Android runtime sweep for exported EfficientFormer artifacts. AUC is computed from exported predictions under the fixed TN5000 test protocol. Hot latency is end-to-end total latency in milliseconds.

Device	Artifact	Runtime	Android AUC	Hot total ms	Hot inference ms	P95 total ms
Xiaomi 24129PN74C	ECA+KD	CPU default	0.9552	31.312	30.024	34.023
Xiaomi 24129PN74C	ECA+KD	CPU-4T	0.9552	31.574	30.190	33.191
Xiaomi 24129PN74C	ECA+KD	XNNPACK-4T	0.9552	42.403	40.859	48.883
Xiaomi 24129PN74C	ECA+KD	NNAPI	0.9552	159.080	157.471	179.902
Samsung SM-X800	ECA+KD	CPU default	0.9552	56.507	55.180	58.360
Samsung SM-X800	ECA+KD	CPU-4T	0.9552	56.417	55.090	58.506
Samsung SM-X800	ECA+KD	XNNPACK-4T	0.9552	107.141	105.773	112.055
Samsung SM-X800	ECA+KD	NNAPI	0.9552	279.982	278.047	318.094

Table 7: Exported mobile-student comparison under CPU default runtime. Offline metrics are from the strict export summary; Android AUC and latency are from the two-device headless run.

Model	Offline AUC	Offline BalAcc	Android AUC	Xiaomi hot ms	Samsung hot ms	Threshold
EfficientFormer-L1+ECA+KD	0.9563	0.8792	0.9552	31.312	56.507	0.87
EfficientFormer-L1+KD	0.9522	0.8699	0.9511	40.364	56.477	0.83
EfficientFormer-L1+ECA	0.9305	0.8400	0.9278	40.517	60.254	0.49

Table 8: EfficientFormer-L1 multi-organ probe. Values are mean ± standard deviation over three seeds.

Protocol	Test domain	AUC	BalAcc
Joint multi-organ	TN5000	0.9212 ± 0.0065	0.8420 ± 0.0044
Joint multi-organ	BUSI	0.8324 ± 0.0192	0.7553 ± 0.0234
Joint multi-organ	AUL	0.7984 ± 0.0367	0.7318 ± 0.0262
LODO zero-shot	TN5000	0.6261 ± 0.0557	0.5723 ± 0.0239
LODO zero-shot	BUSI	0.7339 ± 0.0428	0.6422 ± 0.0415
LODO zero-shot	AUL	0.5936 ± 0.0344	0.5356 ± 0.0572

conflict images can inflate or distort small public ultrasound benchmarks. The audit below is descriptive; it does not change the main benchmark denominator.

Table 9: Descriptive BUSI duplicate and near-duplicate audit.

Rule	Exact groups	Near-duplicate pairs	Label-conflict pairs	Audit artifact
Strict (≤ 2)	1	88	26	`busi_duplicate_audit_strict.csv`
Relaxed (≤ 5)	1	141	41	`busi_duplicate_audit_relaxed.csv`

S8. Removed Legacy Tables

This submission package intentionally removes earlier supplementary tables for CUDA trade-off, non-final Android baselines, BUSI/AUL auto-ROI classification, and calibration. Those tables were useful during development, but they were produced before the final reporting discipline was enforced or before the Android threshold loader was fixed. They are not used as manuscript evidence. They should only be restored after regeneration from case-level prediction files tied to the same fixed split/protocol metadata as the main manuscript.